1	$y = 2 + 10x^{\frac{1}{2}} - 2x^{\frac{3}{2}}$	
(a)	$\frac{\mathrm{d}y}{\mathrm{d}x} = 5x^{-\frac{1}{2}} - 3x^{\frac{1}{2}}$	M1 A1 A1
		(3)
(b)	$x = 2 \Rightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{5}{\sqrt{2}} - 3\sqrt{2}$	M1
	$\frac{5}{\sqrt{2}} - 3\sqrt{2} = \frac{5}{2}\sqrt{2} - 3\sqrt{2} = -\frac{1}{2}\sqrt{2}$	A1
		(2)
		(5 marks)

2(a)	$P(-3,7), Q(9,11)$ and $R(12,2)$	
Way 1	grad $PQ = \frac{11-7}{9--3} = \frac{1}{3}$, grad $QR = \frac{11-2}{9-12} = -3$	M1 A1
	$\frac{1}{3} \times -3 = -1$ so angle $PQR =$ 90°	A1
		(3)
Way 2	$PQ^2 = (9--3)^2 + (11-7)^2 = 160$ $QR^2 = (12-9)^2 + (2-11)^2 = 90$ $PR^2 = (12--3)^2 + (2-7)^2 = 250$	M1 A1
	$PQ^2 + QR^2 = PR^2$ (or e.g. 90 + 160 = 250) so angle $PQR =$ 90° or e.g. $\cos\theta = \frac{160+90-250}{2\sqrt{160}\sqrt{90}} = 0 \Rightarrow \theta = 90°$	A1
		(3)
Way 3	$\overrightarrow{PQ} = \begin{pmatrix} 9--3 \\ 11-7 \end{pmatrix} = \begin{pmatrix} 12 \\ 4 \end{pmatrix}$, $\overrightarrow{QR} = \begin{pmatrix} 12-9 \\ 2-11 \end{pmatrix} = \begin{pmatrix} 3 \\ -9 \end{pmatrix}$	M1A1
	$\overrightarrow{PQ} \bullet \overrightarrow{QR} = \begin{pmatrix} 12 \\ 4 \end{pmatrix} \bullet \begin{pmatrix} 3 \\ -9 \end{pmatrix} = 36-36 = 0$ so angle $PQR =$ 90°	A1
		(3)

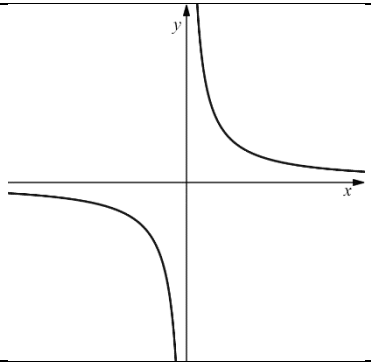
(b)	E.g. $(-3,7)+(3,-9)=...$ or $(12,2)-(12,4)=...$	M1
	$(0,-2)$	A1
		(2)
ALT 1	grad $PQ=\dfrac{11-7}{9--3}=\dfrac{1}{3}\Rightarrow$ eqn RS is $y-2=\dfrac{1}{3}(x-12)$ grad $QR=\dfrac{11-2}{9-12}=-3\Rightarrow$ eqn PS is $y-7=-3(x+3)$ $\Rightarrow x=...,\ y=...$	M1
	$(0,-2)$	A1
ALT 2	Midpoint PR is $\left(\dfrac{9}{2},\dfrac{9}{2}\right)\Rightarrow \dfrac{9+x}{2}=\dfrac{9}{2},\ \dfrac{11+y}{2}=\dfrac{9}{2}\Rightarrow x=..,y=...$	M1
	$(0,-2)$	A1
		(5 marks)

3	$\int \frac{4x^5+3}{2x^2}\,\mathrm{d}x = \int 2x^3 + \frac{3}{2}x^{-2}\,\mathrm{d}x$	M1 A1
	$= \frac{1}{2}x^4 - \frac{3}{2}x^{-1} + c$	dM1 A1 A1
		(5)
		(5 marks)

4	$kx^2 + 6kx + 5 = 0$	
	$b^2 - 4ac = (6k)^2 - 4 \times k \times 5$	M1
	$b^2 - 4ac = (6k)^2 - 4 \times k \times 5 ... 0 \Rightarrow k ...$	dM1
	$k < \frac{5}{9}$	A1
	$0 < k < \frac{5}{9}$	A1
		(4 marks)

5(a)	$9^x = p^2, \quad 3^{x+2} = 9p \quad \text{or} \quad 3^{x-1} = \frac{p}{3}$	B1
	$3\times 9^x + 3^{x+2} = 1 + 3^{x-1} \Rightarrow 3p^2 + 3^2 \times p = 1 + \frac{p}{3^1}$	M1
	$9p^2 + 26p - 3 = 0$ via $3p^2 + 9p = 1 + \frac{p}{3}$*	A1*
		(3)
(b)	$9p^2 + 26p - 3 = 0 \Rightarrow (9p-1)(p+3) = 0$	M1
	$3^x = \frac{1}{9}$	A1 M1 in EPEN
	$x = -2$	A1
		(3)
		(6 marks)

6(a)	$A=\frac{1}{2}r^2\theta \Rightarrow 40=\frac{1}{2}r^2\times 2.4\Rightarrow r=...$	M1
	$r=\sqrt{\frac{80}{2.4}}\Rightarrow r=5.77$ (m)	A1
		(2)
(b)	States or uses that angle AOB = awrt 0.37	B1
		(1)
(c)	$\frac{1}{2}\times "5.77"\times 6.25\times \sin "0.37"\ \left(=6.5377...\right)$	M1
	Full method for area of stage = $40+2\times\frac{1}{2}\times 5.77\times 6.25\times\sin 0.37$	dM1
	= 53.1 m^2	A1
		(3)
(d)	$r\theta = "5.77"\times 2.4=\left(13.848\right)$	M1
	$x^2=6.25^2+"5.77"^2-2\times 6.25\times "5.77"\cos "0.37"\qquad \left(x=2.26\right)$	M1
	Full method for perimeter of stage = $12.5+2\times "2.26"+"5.77"\times 2.4$	ddM1
	= 30.9 m	A1
		(4)
		(10 marks)

7(a)		Correct Shape in quadrant 1 or 3	M1
		Fully correct shape and position	A1
			(2)
(b)	E.g. Translate 2 (units) to the right (parallel to the *x*-axis)		B1, B1
			(2)
(c)	When $x=-4 \Rightarrow \frac{6}{x-2}=kx+7 \Rightarrow \frac{6}{-6}=-4k+7 \Rightarrow k=...$		M1
	$k=2$		A1
			(2)
(d)	$\frac{6}{x-2}="2"x+7 \Rightarrow 6=2x^2+3x-14$		M1
	$0=2x^2+3x-20 \Rightarrow (2x-5)(x+4)=0$		dM1
	$Q=\left(\frac{5}{2},12\right)$		A1, A1
			(4)
			(10 marks)

8(a)	$y = 3x + "c"$ or $y = "m"x - 12'$	M1
	$y = 3x - 12$	A1
		(2)
(b)	$k = 10$	B1
		(1)
(c)	E.g. $y = A(x-4)(x-10)$ or $y = C(x-7)^2 - 18$	M1
	E.g. $-18 = A(7-4)(7-10) \Rightarrow A = ...$ Or $0 = C(4-7)^2 - 18 \Rightarrow C = ...$	dM1
	$y = 2(x-4)(x-10)$, $y = 2(x-7)^2 - 18$ o.e.	A1
		(3)
(d)	Two of $y > 3x - 12$, $y < 2(x-4)(x-10)$, $x > 0$, $x < 4$	M1
	E.g. $3x - 12 < y < 2(x-4)(x-10)$, $0 < x < 4$	A1
		(2)
		(8 marks)

9(a)	π	B1
		(1)
(b)(i)	3	B1
		(1)
(ii)	5	B1
		(1)
(iii)	201	B1
		(1)
		(4 marks)

10(a)	One of $-\frac{20}{3} < x < -6, \quad x > \frac{3}{2}$	M1
	Both $-\frac{20}{3} < x < -6, \quad x > \frac{3}{2}$	A1
		(2)
(b)	$(3x+20)(x+6)(2x-3) = (3x+20)\left(2x^2+9x-18\right) =$	M1
	$= 6x^3 + 67x^2 + 126x - 360$	A1 A1
		(3)
(c)	$\frac{\mathrm{d}y}{\mathrm{d}x} = 18x^2 + 134x + 126 \Rightarrow$ Gradient at $x = 0$ is 126	M1
	Equation of l is $y = 126x - 360$	A1ft
	l cuts C again when $6x^3 + 67x^2 + 126x - 360 = 126x - 360$	dM1
	$6x^3 + 67x^2 = 0 \Rightarrow x^2(6x+67) = 0$	ddM1
	$x = -\frac{67}{6}$	A1
		(5)
		(10 marks)

11(a)	States or implies that the gradient of the normal is $-\frac{1}{3}$	M1
	Correct equation of normal e.g. $y-16=-\frac{1}{3}(x-4)$	A1
		(2)
(b)	$f''(x)=4x+x^{-\frac{1}{2}} \Rightarrow f'(x)=2x^2+2x^{\frac{1}{2}}+c$	M1
	$x=4, f'(x)=3 \Rightarrow\ 3=32+4+c \Rightarrow c=...(-33)$	dM1
	$f'(x)=2x^2+2x^{\frac{1}{2}}-33$	A1
	$f'(x)=2x^2+2x^{\frac{1}{2}}-33 \Rightarrow f(x)=\frac{2}{3}x^3+\frac{4}{3}x^{\frac{3}{2}}-33x+d$	dM1
	$x=4, f(x)=16 \Rightarrow (f(x)=)\frac{2}{3}x^3+\frac{4}{3}x^{\frac{3}{2}}-33x+\frac{284}{3}$	ddM1A1
		(6)
		(8 marks)